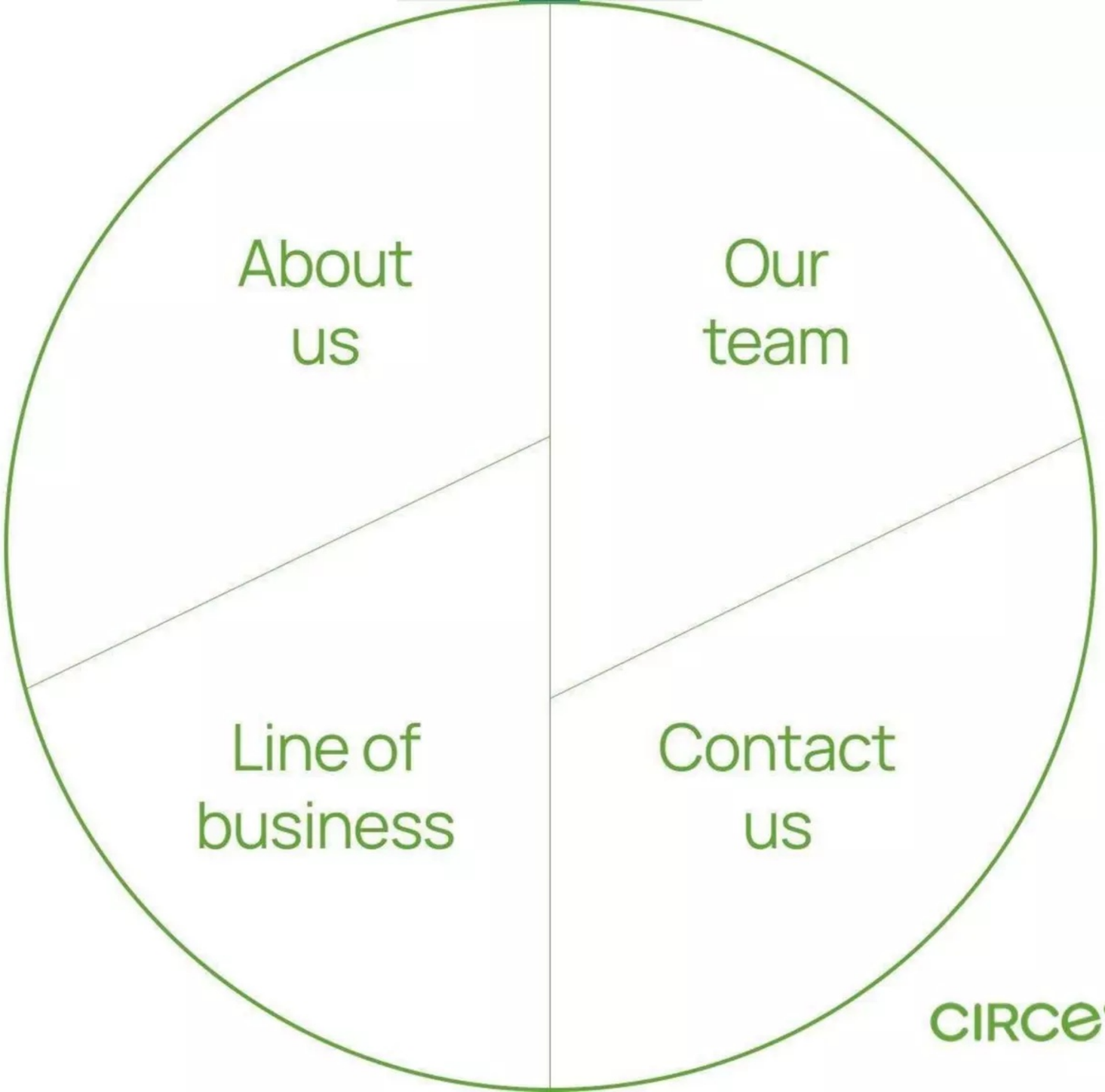


2021 Circe Biotech GmbH

INNOVATIVE TECH IN PRODUCTION
OF LIQUIFIED NATURAL GAS

CIRCE



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About us

2021 Circe Biotech GmbH

CIRCE

Problem



Cattle are the #1 agricultural source of greenhouse gases worldwide.

https://www.ucdavis.edu/food/news/making-cattle-more-sustainable/



The estimate of waste from a broad range of municipal, commercial, and industrial sources is between 7 and 10 billion tonnes per annum.

https://www.unep.org/resources/report/global-waste-management-outlook



Globally, 1 in 9 people is hungry or undernourished.

https://globalnutritionreport.org/reports/2020-global-nutrition-report/

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Mission

Implement new energy-efficient technologies in the present, with concern for the state of the Earth in the future.

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CIRCE

Circe is a group of companies that develops solutions for efficient use of methane.

Our companies conduct a full cycle of work, from R&D to production.

Goal: Annually process

50 bln m^3

of gas into useful products by 2025

CIRCE

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Core values



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2021 Circe Biotech GmbH

Science is at the heart of our work



The company cooperates with leading international and Russian research institutes.

IBPM RAS, SEMENOV ICP,
SRI PCM, PENZSTU, TU WIEN



All achievements are protected by relevant patents and other forms of intellectual property rights.



New technologies go through a full cycle of phases, from the blueprint to the prototype.



We are a resident of the Skolkovo Science and Technology Center.

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CIRCE

Roadmap

FEB 2018

Start of the Bioprotein line

Conducting research

JUNE 2019

Experimental machine creation

Creation of an experimental Bioreactor 1.0

SEP 2019

Experimantal reactor creation

Creation of an experimental reactor for methanol production

MARCH 2020

Manual machine testing

Machine test in manual mode

MAY – JUNE 2020

Agreement with IBPM RAS

Cooperation agreement with the Institute of Biochemistry and Physiology of Microorganisms

JULY 2020

Automatic machine testing

Machine test in automatic mode

Past

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CIRCE

Roadmap

Present

DEC 2020

Agreement with TU Wien

Cooperation agreement with the Technical University of Vienna

DEC 2020

Start of the Biodegradable polymers line

Conducting research

DEC 2020

Agreement with SFAR

Beginning of joint work with the State Fund for Advanced Research

JAN 2021

Skolkovo resident

Resident of Skolkovo Science and Technology Center

JAN 2021

Pilot batch of Bioprotein

Production of a pilot batch of Bioprotein

JUN 2021

Test production

Test production of Vortex tubes

AUG 2021

Methanol 2.0

Creation of methanol reactor 2.0

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CIRCE

Future

Q2 2022
Pilot batch of biodegradable polymers

Q2 2022
Pilot batch of food protein

Q3 2022
Commercial batch of feed protein

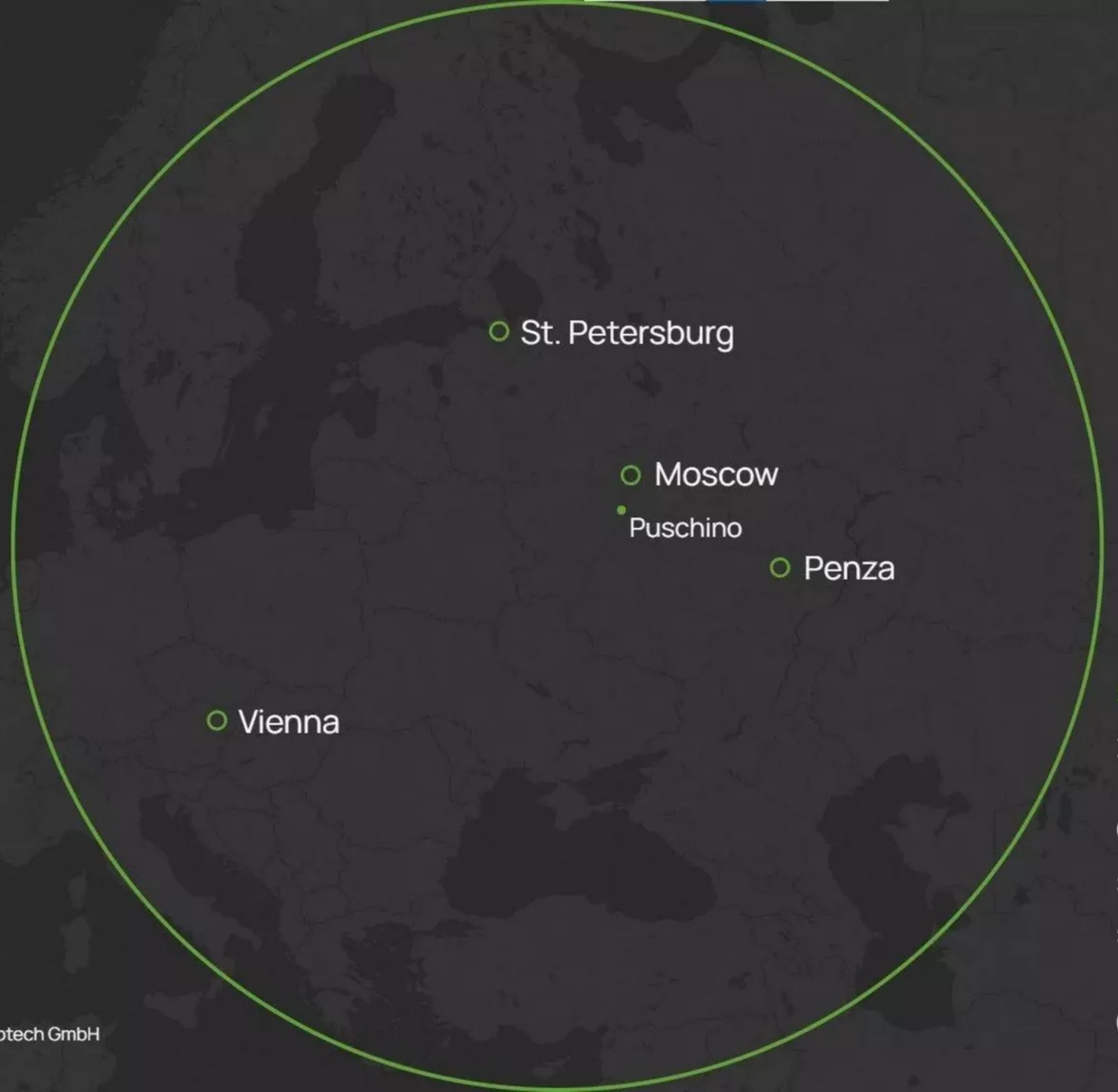
Q2 2023
Commercial batch of food protein

CIRCE°

Our team

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CIRCE



Staff
20 FTE

Outsourcing
25–30 FTE

Circe's staff includes engineers and scientists with proven expertise and scientific backgrounds.

CIRCE

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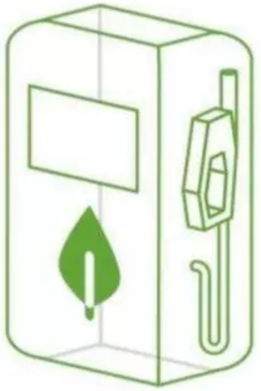
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Business branches



Protein:
feed & food



Methanol



Liquefied
natural gas



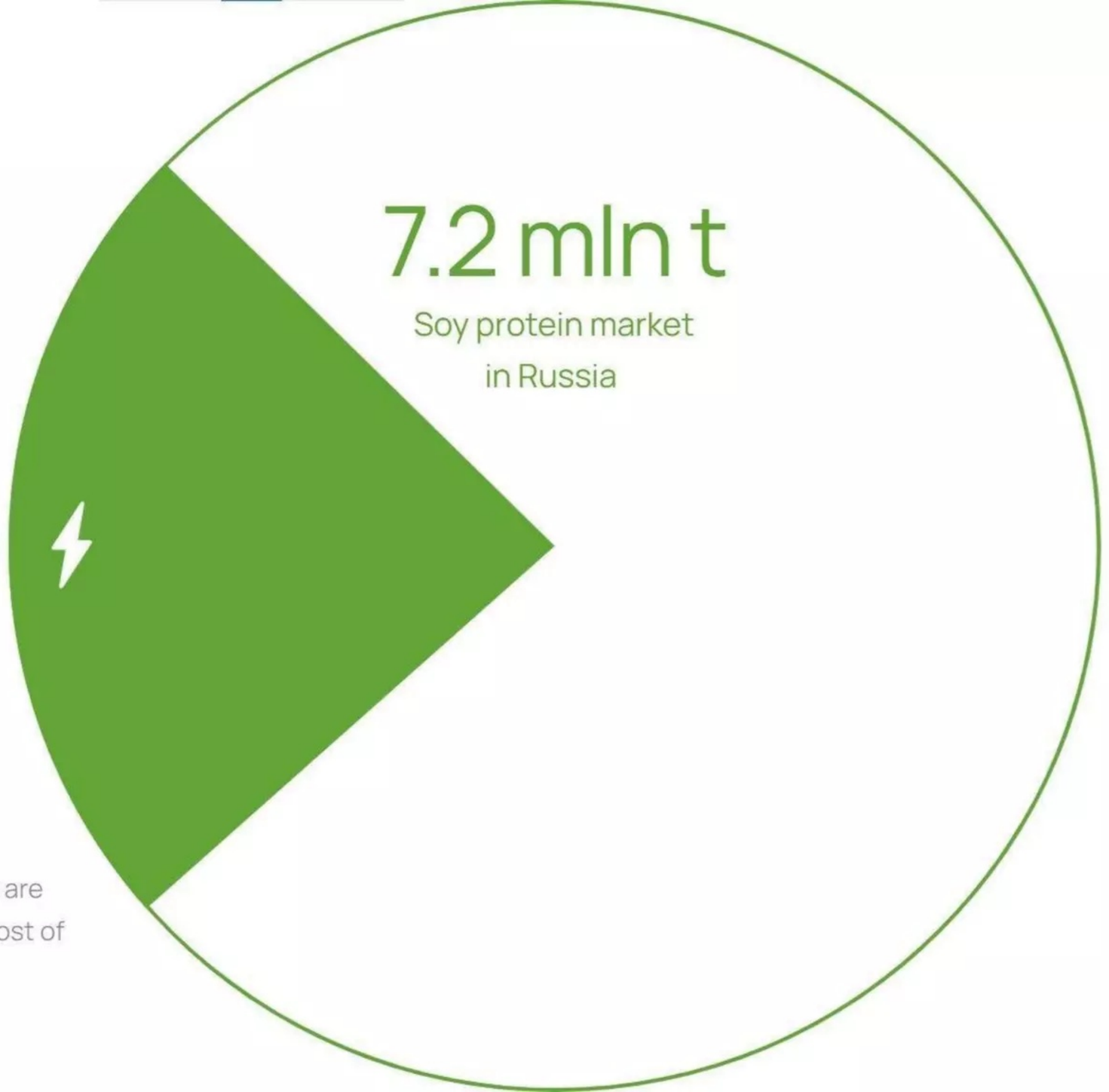
Biodegradable
polymers

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PROTEIN: FEED & FOOD

TARGET MARKET



2-2.5 mln t

Annual protein deficiency

Partially covered by importing*
about 2.2 mln t — equal to
~$1,166 mln in 2020

*The climate conditions in many regions in Russia are not suitable for soybean cultivation, that's why most of the market is represented by imported soybean.

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METHANOL

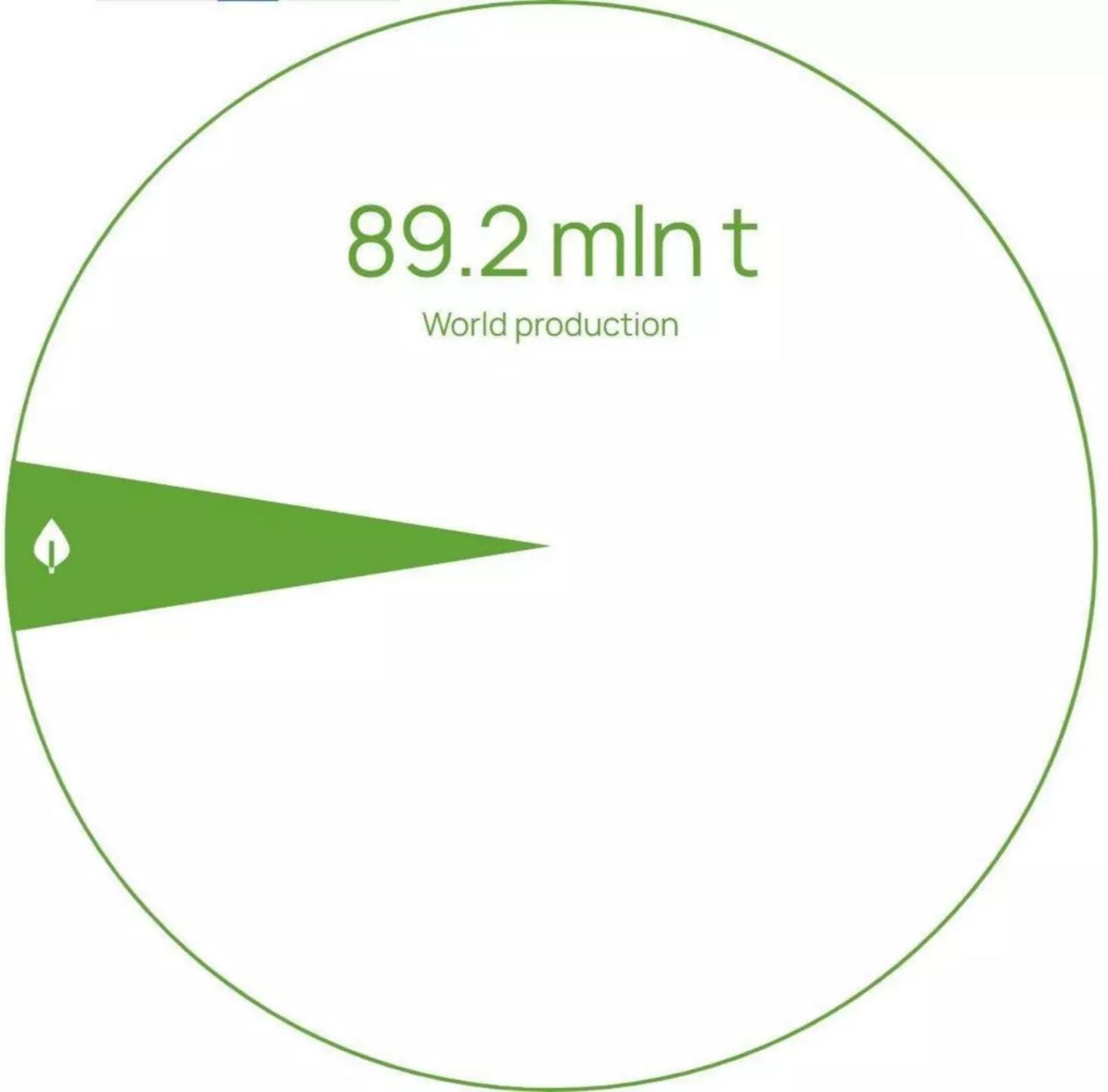
TARGET MARKET

4.46 mln t

Production in Russia

Approximately 5% of total world production

25% annual growth rate



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LIQUEFIED NATURAL GAS

TARGET MARKET

29 mln t

Production in Russia

5.8% annual growth rate



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BIODEGRADABLE POLYMERS

TARGET MARKET

0.01 mln t

Production in Russia

9% annual growth rate



2021 Circe Biotech GmbH



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